



KULDEEP MALIK
DIRECTOR - CORPORATE
SALES INDIA, MEDIATEK

Artificial Intelligence

“By The Upcoming Diwali Season, We May See AI Become An Integral Part Of BUDGET SMARTPHONES”

With artificial intelligence (AI)-integrated chipsets, smartphones are now intelligent and provide unbelievably improved performance, camera quality, battery life and so much more, at reasonable prices. Kuldeep Malik, director - corporate sales India, MediaTek, explains to Paromik Chakraborty of *Electronics For You*, how AI-based chips are transforming the smartphone business

Q. How is AI integration into chipsets improving smartphone performance?

A. Until now, smartphone system-on-chip (SoC) mainly constituted a GPU and a CPU. Majority of AI programs were on the cloud. SoC providers like us are bringing AI into Edge devices. A dedicated unit is required to run AI operations. This is where the AI processing unit (APU) comes in. Chipsets like Helio P60 and P22 have integrated APUs along with GPUs and CPUs in their processors.

An APU takes the load off the GPU and CPU to run AI algorithms. GPU consumes the most power, depleting the phone's battery, while APU consumes the least. Switching over AI tasks to APU drastically enhances battery performance. For instance, Realme 1.0, having a battery of 3400mAh, has achieved performance consistently equivalent to 4000mAh battery. Since the programs are handled by a separate dedicated unit, the device responds much quicker to commands, improving speed and performance.

Q. What other parameters are impacted by APUs?

A. Through neural network driven engines, AI devices learn and behave in a way that suits the users' usage patterns and preferences. AI fine-tunes the various parameters of the devices, including GPU and CPU, to go into optimal usage mode.

For example, AI identifies and groups photos of one person in one place. It also analyses videos on-the-

fly and triggers alarms or notifications immediately if it identifies suspicious events.

Q. How are AI-driven chips changing the smartphone business?

A. Earlier, AI was expected only in high-range smartphones. We want to bring AI across all price ranges. The notion is already changing, case in point being Realme 1.0, which is available at ₹ 14,000. The ₹ 7000 to ₹ 10,000 price range is lucrative, for both buyers and sellers. Hence, we are trying to bring advanced technologies into this price range, which we call the new premium. In fact, by the upcoming Diwali season, we may see AI become an integral part of budget smartphones.

Q. How can developers leverage this intelligence brought by chips?

A. Partners like Google and Facebook are bringing machine learning and AI tools for developers—also accessible through MediaTek platforms. With these tools, we expect a good amount of intelligence being added to non-APU SoCs for entry-level devices as well.

Through our NeuroPilot Engine, we provide a software development kit to developers. Companies like FaceTime and Face++ have already developed their solutions using the same.

Q. How will 5G complement these intelligent applications?

A. 5G is about high data speed and reduced latency. This will complement various life-impacting smart applica-

tions. For example, autonomous cars and advanced driver-assistance systems will require dedicated bandwidth for consistent and safe functioning.

5G will also drive the popularity of 4K content streaming by improving speed and responses.

Narrow-band Internet of Things (NB IoT) will find more applications as well.

Q. How is MediaTek preparing for it?

A. We are in close collaborations with equipment vendors like Ericsson, Nokia and Huawei. In Europe, we are performing tests with Nokia.

Our first 5G solution, M70, expected to be launched in 2019, is a thin modem that can be used for most smart applications. For example, digital surveillance cameras are prone to damage and theft. With 5G thin modems, data will be sent directly to central storage, ensuring that it is safe even if the cameras are tampered with.

Q. What are your projections for India's smartphone manufacturing ecosystem?

A. India's hardware ecosystem is maturing. Smartphones are being assembled here, and many companies are using SMT lines to manufacture PCBAs here as well.

Gradually, component players will also start setting up their plants in India. However, the pace at which this happens will depend on government policies and the willingness of component makers to create in India. **EFY**